[PROJECT TITLE LINE 1]
[Project Title Line 2 / Subtitle]

By

[Student 1 Full Name]
Student ID: [Student 1 ID]

[Student 2 Full Name]
Student ID: [Student 2 ID]

FINAL YEAR DESIGN PROJECT REPORT

This Report Presented in Partial Fulfillment of the Requirements for the **Degree of Bachelor of Science in Computer Science and Engineering**

Supervised by

[Supervisor Full Name]
[Supervisor’s Academic Designation]
Department of Computer Science and
Engineering Daffodil International University

Co-Supervised by

[Co-Supervisor Full Name]
[Co-Supervisor’s Academic Designation]
Department of Computer Science and
Engineering Daffodil International University

DAFFODIL INTERNATIONAL UNIVERSITY
Dhaka, Bangladesh
[Submission Month, Year]

APPROVAL

This Project titled "**[PROJECT TITLE]**," submitted by **[Student 1 Name]** (ID: [Student 1 ID]) and **[Student 2 Name]** (ID: [Student 2 ID]) to the Department of Computer Science and Engineering, Daffodil International University, has been accepted as satisfactory for the partial fulfillment of the requirements for the degree of B.Sc. in Computer Science and Engineering and approved as to its style and contents.

BOARD OF EXAMINERS

[Board Chairman Name] Board Chairman Designation, Department of CSE, FSIT Daffodil International University	.... Signature
[Internal Examiner 1 Name] Internal Examiner 1 Designation, Department of CSE, FSIT Daffodil International University	.... Signature
[Internal Examiner 2 Name] Internal Examiner 2 Designation, Department of CSE, FSIT Daffodil International University	.... Signature
[External Examiner Name] External Examiner Designation, Department of CSE, FSIT Daffodil International University	.... Signature

DECLARATION

We hereby declare that this project report titled **[PROJECT TITLE]** has been carried out by us under the supervision of **[Supervisor Name]**, [Supervisor Designation], Department of Computer Science and Engineering, Daffodil International University. We further declare that neither this project nor any part thereof has been submitted elsewhere for the award of any academic degree, diploma, or certificate.

Supervised by:

. . . .

[Supervisor Name]
[Supervisor Designation]
Department of CSE, FSIT
Daffodil International University

Co-Supervised by:

. . . .

[Co-Supervisor Name]
[Co-Supervisor Designation]
Department of CSE, FSIT
Daffodil International University

Submitted by:

. . . .

[Student 1 Name]
Student ID: [Student 1 ID]
Department of CSE, FSIT
Daffodil International University

. . . .

[Student 2 Name]
Student ID: [Student 2 ID]
Department of CSE, FSIT
Daffodil International University

ACKNOWLEDGEMENTS

We express our deepest gratitude to the Almighty for granting us the capability, perseverance, and determination to complete this Final Year Design Project (FYDP) successfully.

We express our sincere indebtedness and gratitude to our respected supervisor, **[Supervisor Name]**, for their continuous guidance, scholarly advice, and invaluable encouragement throughout every stage of this research.

We also thank the Head of the Department of Computer Science and Engineering, faculty members, and lab officers for their institutional support and technical facilities.

Finally, we express our profound appreciation to our parents and family members for their constant prayers, patience, and unconditional sacrifices.

ABSTRACT

[Write a concise, 250–350 word structured abstract summarizing: (1) Background context and problem motivation, (2) Core research gap and proposed methodology/system architecture, (3) Key algorithms, technologies, and implementation highlights, and (4) Quantitative benchmark results and significance of the contributions.]

Contents

List of Figures

List of Tables

Chapter 1

Introduction

1.1 Background and Context

[Provide the high-level background of your domain. Introduce the broader field (e.g., Intelligent Transportation Systems, Healthcare IoT, Distributed Systems, Computer Vision, Cybersecurity) and establish the practical context of the problem within Bangladesh or global scenarios.]

1.2 Problem Statement

[Clearly describe the specific problem, inefficiency, or research challenge your project solves. Contrast current manual or suboptimal approaches with what is needed.]

1.3 Motivation and Significance

[Explain why this problem is critical to solve. Detail computational, technological, economic, and societal motivations.]

1.4 Project Objectives

The primary objectives of this project are:

1. **[Objective 1]**: [Describe the primary hardware/software/algorithmic objective].
2. **[Objective 2]**: [Describe the secondary system architecture/data processing objective].
3. **[Objective 3]**: [Describe the analytical, ML, or algorithmic evaluation objective].

4. **[Objective 4]**: [Describe the user interface, deployment, or integration objective].

1.5 Scope and Delimitations

- **In Scope**: [List features, models, protocols, and boundaries included in this work].
- **Delimitations / Exclusions**: [Explicitly declare what is outside the current scope or reserved for future work].

1.6 Research Methodology Overview

[Summarize the engineering and research lifecycle used: requirements gathering $\rightarrow$ architectural design $\rightarrow$ hardware/software implementation $\rightarrow$ empirical evaluation and benchmarking.]

1.7 Organization of the Report

The remainder of this report is structured as follows:

- **Chapter 2 (Literature Review and Background)**: Reviews theoretical foundations, related academic literature, and feature gap matrices.
- **Chapter 3 (Requirements Analysis and Feasibility)**: Analyzes stakeholder needs, user personas, functional and non-functional specifications.
- **Chapter 4 (System Architecture and Design)**: Details system blueprints, hardware BOM, network topology, DFDs, and database ERD schemas.
- **Chapter 5 (System Implementation)**: Explains embedded firmware, core algorithms, microservices, and full-stack web/mobile interfaces.
- **Chapter 6 (Testing, Results, and Evaluation)**: Documents verification methodologies, benchmark performance, confusion matrices, and latency measurements.
- **Chapter 7 (Engineering Standards and CEP Mappings)**: Maps compliance with international standards, societal/environmental sustainability, and BAETE EP1–EP7, K1–K8, EA1–EA5 mappings.

- **Chapter 8 (Conclusion and Future Work)**: Summarizes contributions, operational limitations, and future research directions.

Chapter 2

Background and Literature Review

2.1 Introduction

[Introduce the theoretical domain, core scientific concepts, and the structure of your literature review.]

2.2 Theoretical Foundations

2.2.1 [Theoretical Concept 1: Core Mathematical / Scientific Model]

[Provide the mathematical formulations, equations, and theoretical definitions relevant to your domain.]

$$\mathbf{y} = f(\mathbf{x}, \theta) + \epsilon \tag{2.1}$$

2.2.2 [Theoretical Concept 2: Algorithmic Foundations]

[Detail algorithms, data structures, communication protocols, or neural network architectures employed.]

2.3 Systematic Literature Review

[Provide a comprehensive review of peer-reviewed literature in your domain.]

Table 2.1: Systematic Summary of Contemporary Literature Reviewed

Author(s)	**Year**	**Title / Domain**	**Methodology**	**Identified Limitations / Gaps**
[Author et al.]	[Year]	[Paper Title 1]	[Methodology Used]	[Identified gap or limitation in this approach].
[Author et al.]	[Year]	[Paper Title 2]	[Methodology Used]	[Identified gap or limitation in this approach].
[Author et al.]	[Year]	[Paper Title 3]	[Methodology Used]	[Identified gap or limitation in this approach].

2.4 Comparative Analysis of Existing Systems

[Discuss commercial applications, open-source projects, or municipal systems currently deployed in this domain.]

2.5 Research Gap Analysis

Table 2.2: System Feature Gap Analysis Matrix

Features / Architectural Capabilities	**[System A]**	**[System B]**	**[System C]**	**Proposed System**
[Feature / Capability 1]	Yes	No	Partial	**Yes**
[Feature / Capability 2]	Partial	No	No	**Yes**
[Feature / Capability 3]	No	No	No	**Yes**
[Feature / Capability 4]	Yes	No	Yes	**Yes**

2.6 Summary

[Summarize how the proposed project addresses the identified gaps].

Chapter 3

Research Methodology and System Design

Every chapter should start with 1-2 sentences on the outline of the chapter. [This chapter outlines the research methodology, stakeholder analysis, functional and non-functional requirements, system architecture, hardware BOM, network design, and relational database models.]

3.1 Research Methodology

[Detail the research lifecycle, phases, iterative development workflow, and empirical validation plan.]

3.2 Stakeholder and Requirements Analysis

3.2.1 Stakeholder Analysis

Table 3.1: Stakeholder Analysis and System Needs Mapping

Stakeholder Group	**Core Role / Context**	**System Needs & Expectations**
[Stakeholder 1]	[Primary role]	[Core requirements expected from system].
[Stakeholder 2]	[Secondary role]	[Core requirements expected from system].
[End Users]	[Direct users]	[User experience and operational needs].

3.2.2 Functional Requirements Specification

Table 3.2: Functional Requirements Specification (FRS) Matrix

Req ID	Requirement Title	Functional Description	Priority
FR-01	[Data Ingestion / Capture]	[Detailed description of functional behavior].	High
FR-02	[Core Processing / Algorithm]	[Detailed description of functional behavior].	High
FR-03	[Storage / Database]	[Detailed description of functional behavior].	High
FR-04	[User Interface / Alerts]	[Detailed description of functional behavior].	Medium

3.3 System Architecture and Component Design

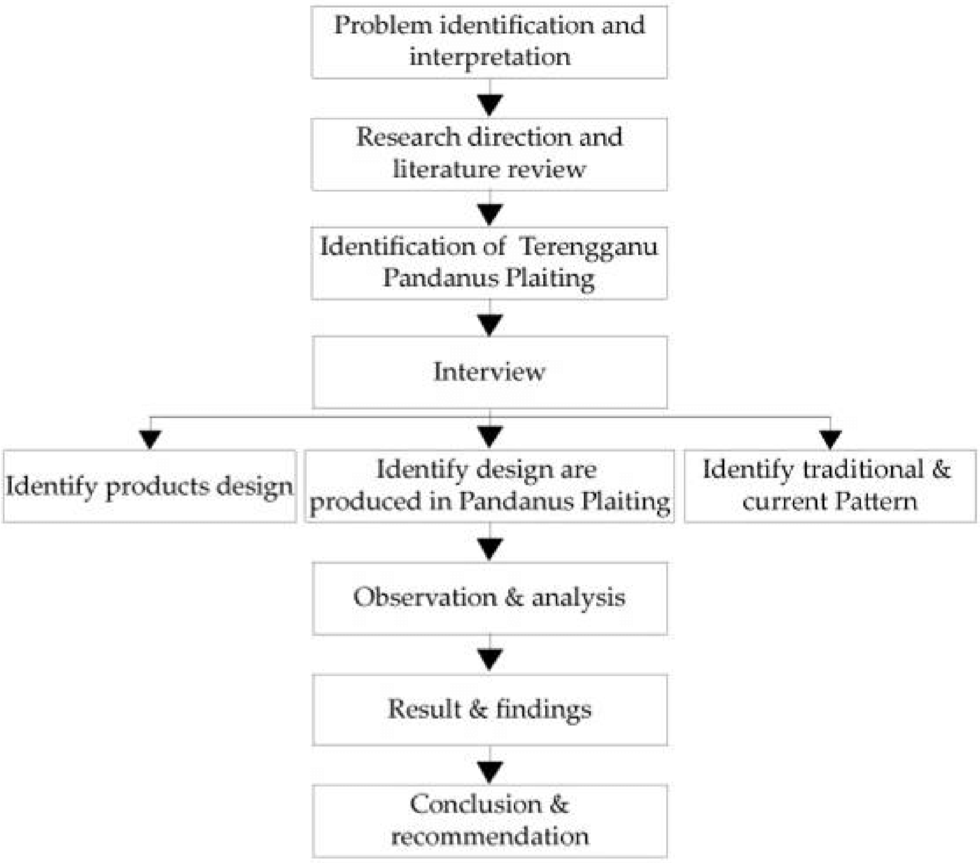

Figure 3.1: High-Level System Architecture Diagram

3.4 Hardware Architecture and Bill of Materials (BOM)

Table 3.3: Hardware Bill of Materials (BOM) and Cost Breakdown

Component	**Model / Specifications**	**Operational Role**	**Est. Cost (BDT)**
[Microcontroller]	[e.g., ESP32 / Arduino / Raspberry Pi]	[Sensor reading and networking]	[Cost]
[Primary Sensor / Camera]	[Model / Part Number]	[Data collection / vision capture]	[Cost]
[Power / Enclosure]	[Regulator / Battery / Case]	[Power regulation and protection]	[Cost]
Total Hardware Prototyping Cost			**[Total BDT]**

3.5 Data Flow Modeling (DFD)

3.5.1 Context Diagram (DFD Level 0)

[Explain external entities, inputs, and outputs.]

3.5.2 Decomposed Data Flow Diagram (DFD Level 1)

[Detail sub-processes, data stores, and internal data flows.]

3.6 Database Design and Relational Schema (ERD)

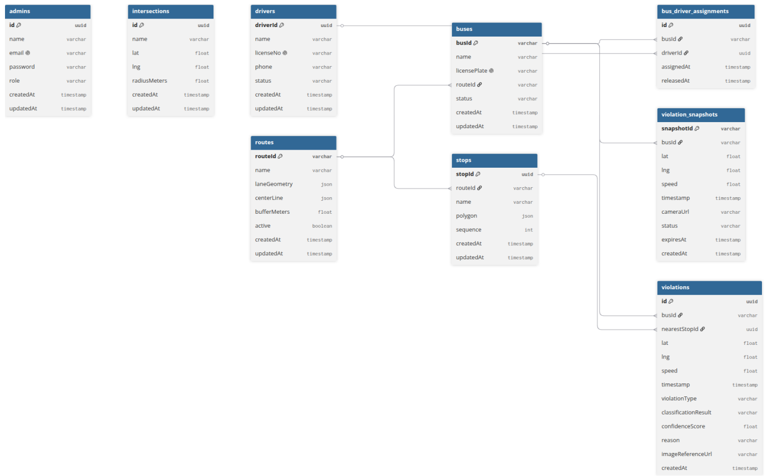

Figure 3.2: Relational Entity-Relationship Diagram (ERD)

Chapter 4

Implementation, Testing, and Results

Every chapter should start with 1-2 sentences on the outline of the chapter. [This chapter describes the development environment, embedded and software implementations, core algorithms, API designs, experimental testing frameworks, and quantitative performance benchmark results.]

4.1 Implementation Environment and Tools

- **Hardware / Firmware**: [e.g., C++, FreeRTOS, Arduino IDE / PlatformIO].
- **Backend / Microservices**: [e.g., Node.js / Express, Python / FastAPI, Django].
- **Database & Storage**: [e.g., PostgreSQL, Prisma ORM, Redis, Firebase].
- **Frontend / Client**: [e.g., Next.js 16, React 19, Flutter].

4.2 Core Algorithmic Implementation

4.2.1 [Core Algorithm / Estimation Pipeline]

```
def process_data_pipeline(data_payload):
    # Step 1: Ingest and validate data
    validated = validate_payload(data_payload)

    # Step 2: Execute algorithmic estimation / filtering
    result = compute_estimation(validated)

    return result
```

4.3 Experimental Testing, Benchmarks, and Results

4.3.1 Experimental Performance Benchmarks

Table 4.1: [Algorithmic / System Performance Benchmark Results]

Test Parameter / Metric	**Baseline / Traditional**	**Proposed System**	**Quantitative Improvement**
[Metric 1: Latency / Error]	[Baseline Value]	[Proposed Value]	**[X% Improvement]**
[Metric 2: Throughput / Speed]	[Baseline Value]	[Proposed Value]	**[X% Improvement]**
[Metric 3: Success / Accuracy Rate]	[Baseline Value]	[Proposed Value]	**[X% Improvement]**

4.3.2 Machine Learning Classification Performance

Table 4.2: [Model Evaluation Metrics Across Target Classes]

Evaluation Metric	**[Class A]**	**[Class B]**	**[Class C]**	**Macro Average**
Precision	0.94	0.91	0.95	**0.93**
Recall	0.93	0.89	0.93	**0.92**
F1-Score	0.93	0.90	0.94	**0.93**
Overall Accuracy			**[94.2%]**	
Inference Latency			**[68 ms]**	

4.4 Discussion of Results

[Discuss key takeaways, statistical significance, and operational performance implications.]

Chapter 5

Engineering Standards and Design Challenges

[Must be present in Phase-1 Progress Report and also in Final Report] Every chapter should start with 1-2 sentences on the outline of the chapter.

5.1 Compliance with the Standards

Only mention the standards that are related to your project. This list is not complete. For each of the standards discuss the alternates with pros and cons and rationale of selection.

5.1.1 Software Standards

[Discuss selected software engineering standards (e.g., ISO/IEC 25010, RFC 7946 GeoJSON, REST RFC 7231, W3C standards) alongside evaluated alternatives, their pros and cons, and your selection rationale.]

5.1.2 Hardware Standards

[Discuss selected hardware standards (e.g., ISO 16750-2 automotive electrical safety, IEEE 802.3 Ethernet, IEEE 802.11 WiFi, RoHS) alongside evaluated alternatives, pros, cons, and selection rationale.]

5.1.3 Communication Standards

[Discuss selected communication and security protocols (e.g., RFC 6455 WebSockets, TLS 1.3, MQTT, Bluetooth SIG) alongside evaluated alternatives, pros, cons, and selection rationale.]

5.2 Impact on Society, Environment and Sustainability

5.2.1 Impact on Life

[Discuss how the proposed project impacts human life, health, convenience, public safety, and daily living conditions.]

5.2.2 Impact on Society & Environment

[Discuss broader societal implications, urban benefits, environmental sustainability, energy conservation, and carbon footprint reduction.]

5.2.3 Ethical Aspects

[Discuss ethical considerations, data privacy, prevention of bias, intellectual property, and user consent management.]

5.2.4 Sustainability Plan

[Explain how the system will remain operational, maintainable, durable, and scalable over the long term.]

5.3 Project Management and Financial Analysis

Provide a cost analysis in terms of budget required and revenue model. In case of budget, you must show an alternate budget and rationales.

Table 5.1: Proposed Budget vs. Alternate Budget Analysis

Cost Component / Item	**Proposed Budget**	**Alternate Budget**	**Selection Rationale**
[Hardware Modules / Sensors]	[Cost A]	[Cost B]	[Rationale for selecting proposed components over commercial alternatives].
[Cloud Hosting / Backend]	[Cost A]	[Cost B]	[Rationale for server/cloud infrastructure].
[Networking / Telematics]	[Cost A]	[Cost B]	[Rationale for connectivity plans].
Total Development Budget	**[Total A]**	**[Total B]**	**[Quantitative Savings Rationale]**

5.4 Complex Engineering Problem

5.4.1 Complex Problem Solving

In this section, provide a mapping with problem solving categories. For each mapping add subsections to put rationale (Use Table 5.1). For P1, you need to put another mapping with Knowledge profile and rational thereof.

Table 5.2: Mapping with Complex Engineering Problem (Table 5.1)

EP1	**EP2**	**EP3**	**EP4**	**EP5**	**EP6**	**EP7**
Dept of Knowledge	Range Of Conflicting Requirements	Depth of Analysis	Familiarity of Issues	Extent of Applicable Codes	Extent Of Stakeholder Involvement	Interdependence
✓	✓	✓	✓	✓	✓	✓

Rationale for EP1 (Depth of Knowledge)

[Explain why depth of engineering knowledge across multiple disciplines is required to solve this problem.]

Rationale for EP2 (Range of Conflicting Requirements)

[Explain conflicting constraints such as speed vs. memory, accuracy vs. latency, or cost vs. durability.]

Rationale for EP3 (Depth of Analysis)

[Detail mathematical modeling, signal/data processing, or algorithmic complexity required.]

Rationale for EP4 (Familiarity of Issues)

[Explain non-standard, ill-structured, or novel challenges in this domain.]

Rationale for EP5 (Extent of Applicable Codes)

[List engineering codes, international safety, and communication standards enforced.]

Rationale for EP6 (Extent of Stakeholder Involvement)

[Analyze interactions with diverse stakeholders with conflicting priorities.]

Rationale for EP7 (Interdependence)

[Explain high interdependence between embedded IoT, cloud, AI models, and user dashboards.]

5.4.2 Mapping with Knowledge Profile

This section is designed to map the overall problem and EP1 (*multiple between K3, K4, K5, K6, K8 for attaining EP1*) to the Knowledge Profile.

Table 5.3: Mapping with Knowledge Profile (Table 5.2)

K1	K2	K3	K4	K5	K6	K7	K8
Natural Science	Mathematics	Engineering Fundamentals	Specialist Knowledge	Engineering Design	Engineering Practice	Comprehension	Research Literature
✓	✓	✓	✓	✓	✓	✓	✓

Rationale for Knowledge Profile (K1–K8)

- **K1 (Natural Science)**: [Applied physical principles, optical properties, or circuit physics].
- **K2 (Mathematics)**: [Applied linear algebra, probability, statistics, and discrete math].
- **K3 (Engineering Fundamentals)**: [Core CS/EE foundations, data structures, and computer organization].
- **K4 (Specialist Knowledge)**: [Advanced domain concepts in AI, IoT, Networks, or Distributed Systems].
- **K5 (Engineering Design)**: [Systematic design process from requirements to full implementation].
- **K6 (Engineering Practice)**: [Use of industry tools, IDEs, version control, and CI/CD pipelines].
- **K7 (Comprehension)**: [Understanding of societal, environmental, and legal contexts].
- **K8 (Research Literature)**: [Systematic review and citation of peer-reviewed literature].

5.4.3 Engineering Activities

In this section, provide a mapping with engineering activities. For each mapping add subsections to put rationale (Use Table 5.3).

Mapping with Complex Engineering Activities

This section is designed to map the overall problem and EA's (*multiple*).

Table 5.4: Mapping with Complex Engineering Activities (Table 5.3)

EA1	**EA2**	**EA3**	**EA4**	**EA5**
Range of resources	Level of Interaction	Innovation	Consequences for society and environment	Familiarity
✓	✓	✓	✓	✓

Rationale for Complex Engineering Activities (EA1–EA5)

- **EA1 (Range of Resources)**: [Coordination across hardware, microservices, cloud VPS, and databases].
- **EA2 (Level of Interaction)**: [Interaction across multidisciplinary teams, end-users, and administrators].
- **EA3 (Innovation)**: [Novel algorithms, workflows, or hardware configurations formulated].
- **EA4 (Consequences for Society and Environment)**: [Impact on public welfare, safety, and pollution reduction].
- **EA5 (Familiarity)**: [Solving complex challenges under dynamic, uncertain real-world conditions].

5.5 Summary

[Provide a 1–2 paragraph concluding summary of Chapter 5].

Chapter 6

Conclusion

[Must be present in Phase-1 Progress Report and also in the Final Report. Might be incomplete in FYDP-1 Report.] Every chapter should start with 1-2 sentences on the outline of the chapter.

6.1 Summary

[Provide a comprehensive summary of the project, including motivation, developed architecture, key features, and experimental results.]

6.2 Limitation

[Describe the known technical, operational, environmental, or hardware limitations of the current prototype.]

6.3 Future Work

[Detail future extensions, commercialization plans, scaling optimizations, and subsequent research phases.]

Bibliography

[1] Jon Kleinberg and Eva Tardos. *Algorithm design*. Pearson Education India, 2006.

[2] First Author and Second Author. Article title. *Journal Name*, Volume Number(Issue Number):Start Page–End Page, Year of Publication.

[3] John Smith and Jane Doe. The impact of climate change on coastal ecosystems. *Nature Climate Change*, 13(2):123–145, 2023.

[4] Board of Accreditation for Engineering and Technical Education (BAETE). Manual for Accreditation of Undergraduate Engineering Programs in Bangladesh. *Institution of Engineers, Bangladesh (IEB)*, 2nd Edition, Dhaka, Bangladesh, 2023.